

Part 1 of 3

READING THE DATASHEETS OF ELECTRONIC DEVICES

A thorough understanding of datasheets allows the designers to select the right device for their application and subsequently exploit the potential of the device to the fullest. Beginning this month in a series of three articles, we would focus on how to read a datasheet and then go on to discuss the datasheets of two of the most commonly used electronics devices representing discrete devices and integrated circuits

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Most of the present-day electronics designers and application engineers do not find it necessary to look beyond the first page of the datasheets of the electronic components or devices while selecting one for their application. They consider the knowledge of a few major specifications of the device sufficient for the design exercise that they wish to undertake. Though it may be adequate in some cases, the design where the components are chosen without giving due consideration to the information contained in the datasheet may not be the optimal one.

The datasheet of a device reveals its true character and hence its suitability for a given application. In addition to listing down the device parameters, it gives the electrical and environmental conditions at which these parameters have been measured and also contains information on variation of important parameters as a function of other relevant electrical and environmental parameters. It is vital information particularly when you are designing a circuit or a module that has to qualify stringent environment standards. A comprehensive datasheet almost invariably includes application circuits recommended by the manufacturer.

Comprehensive and condensed datasheets

Both comprehensive as well as condensed versions of datasheets are usu-

30A, 200V, 0.085 Ohm, N-Channel Power MOSFET

This N-Channel enhancement mode silicon gate power field effect transistor is designed, tested and guaranteed to withstand a specified level of energy in the breakdown avalanche mode of operation. These MOSFETs are designed for applications such as switching regulators, switching converters, motor drivers, relay drivers, and drivers for high power bipolar switching transistors requiring high speed and low gate drive power. They can be operated directly from integrated circuits. Formerly developmental type TA09295

Ordering Information

PART NUMBER	PACKAGE	BRAND
IRF250	TO-204AE	IRF250

NOTE: When ordering, include the entire part number.

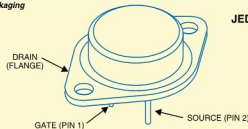
Features

- 30A, 200V
- $r_{DS(on)} = 0.085\Omega$
- Single Pulse Avalanche Energy Rated
- SOA is Power Dissipation Limited
- Nanosecond Switching Speeds
- Linear Transfer Characteristics
- High Input Impedance
- Related Literature
- TB334 Guidelines for Soldering Surface Mount Components to PC Boards

Symbol



Packaging



JEDEC TO-204AE TOP VIEW

Features tell the electrical characteristics, both general and exceptional

Applications give hint to the designers whether the device is suitable for their application or not

Fig. 1: First page of datasheet of power MOSFET type IRF250

ally available from manufacturers. The word 'comprehensive' is used here to differentiate between the datasheet of a device found in a short-form catalogue and another datasheet of the same device available in the detailed version. For example, a company manufacturing a range of semiconductor devices may have one condensed catalogue containing data on all categories of devices like diodes, bipolar transistors, MOSFETs and thyristors that it manufactures. It may also offer detailed catalogues separately for each of these categories.

Surely, the technical data of a given device in the short-form catalogue will not be as comprehensive as it would be in the detailed catalogue. In the condensed version, one is likely to find salient features, major performance specifications, package information and electrical characteristics along with the condition for which they are valid. On the other hand, a detailed datasheet offers a range of characteristic curves showing variation of important parameters as a function of other relevant parameters and typical application circuits.